

ADULT CCRN AT A GLANCE

Owner	AACN Certification Corporation
Items	150 multiple-choice — 125 scored, 25 unscored
Time	3 hours
Framework	AACN Synergy Model for Patient Care
Split	Clinical Judgment 80% · Professional Caring and Ethical Practice 20%
Test plan	Current Adult test plan applies to exams from 12 Nov 2025

ADULT CCRN TEST PLAN

Section	Weight
Clinical Judgment — Cardiovascular	13%
Clinical Judgment — Respiratory	12%
Endocrine, Hematology/Immunology, GI, Renal/GU, Integumentary	21%
Musculoskeletal, Neurological, Behavioral/Psychosocial	18%
Multisystem	16%
Professional Caring and Ethical Practice	20%

TIP Percentages are published rounded and may not total exactly 100. Multisystem plus the combined-system sections outweigh cardiac — do not over-study rhythms alone.

CALCULATIONS WORTH MEMORISING

Mean arterial pressure	$\text{MAP} = (\text{SBP} + 2 \times \text{DBP}) \div 3$ Perfusion target usually ≥ 65 mmHg
Cerebral perfusion pressure	$\text{CPP} = \text{MAP} - \text{ICP}$ Usual goal 60–70 mmHg
Cardiac output	$\text{CO (L/min)} = \text{HR (beats/min)} \times \text{SV (mL/beat)} \div 1000$ $\text{CI} = \text{CO} \div \text{body surface area}$
Anion gap	$\text{Na} - (\text{Cl} + \text{HCO}_3)$ Roughly 8–12 mEq/L
P/F ratio	$\text{PaO}_2 \div \text{FiO}_2$ ≤ 300 on appropriate support is an ARDS oxygenation criterion; diagnosis also needs compatible acute clinical and imaging criteria
Corrected calcium	$\text{Total Ca} + 0.8 \times (4.0 - \text{albumin})$ Or use an ionised calcium
Tidal volume, lung protective	4–8 mL per kg predicted body weight Keep plateau pressure < 30 cmH₂O

CARDIOVASCULAR RECALL

- ST elevation in II, III and aVF is inferior — check right-sided leads.
- V1–V4 is anterior (left anterior descending); I, aVL, V5–V6 is lateral.
- New right bundle branch block with chest pain is an ischaemia red flag.
- Beck triad — hypotension, distended neck veins, muffled heart sounds.

RESPIRATORY AND VENTILATION RECALL

- High peak with normal plateau means resistance: secretions, bronchospasm, a kinked tube.
- High peak with high plateau means compliance: ARDS, oedema, pneumothorax, abdominal pressure.
- Sudden hypotension with absent breath sounds on ventilation suggests tension pneumothorax.
- PEEP recruits alveoli and improves oxygenation but can drop preload.
- Auto-PEEP builds when expiratory time is too short — lengthen it.

MULTISYSTEM RECALL

- Sepsis: take blood cultures as soon as possible, ideally before antibiotics but without delaying them; start antimicrobials promptly and begin fluid resuscitation immediately for hypoperfusion, adding vasopressors if hypotension persists.
- DKA: start fluids and check potassium before insulin. If potassium is below about 3.5 mmol/L, replace it and delay insulin; otherwise start insulin and add potassium as levels fall.
- Hyperosmolar hyperglycaemic state: higher glucose, higher osmolality, little ketosis.
- Rhabdomyolysis: rising creatine kinase, dark urine, watch potassium and kidneys.
- Burns: assess the airway immediately when inhalation injury is suspected and intubate early if compromise is developing, then resuscitate fluids by protocol.



HEMODYNAMICS — ADULT

Parameter	Usual range
Central venous / right atrial pressure	2–6 mmHg
Pulmonary artery pressure	15–30 / 8–15 mmHg
Pulmonary artery occlusion (wedge)	8–12 mmHg
Cardiac output	4–8 L/min
Cardiac index	2.5–4.0 L/min/m ²
Systemic vascular resistance	800–1200 dynes·s·cm ⁻⁵
Stroke volume	60–100 mL
Central venous oxygen saturation (ScvO ₂)	70–80%
Intracranial pressure	5–15 mmHg

ABG AND LABS — REFERENCE RANGES VARY BY LAB

Test	Usual range
pH	7.35–7.45
PaCO ₂	35–45 mmHg
HCO ₃ ⁻	22–26 mEq/L
PaO ₂	80–100 mmHg
Lactate	0.5–2.0 mmol/L
Sodium / potassium	135–145 / 3.5–5.0 mEq/L
Magnesium / phosphate	1.5–2.5 / 2.5–4.5 mg/dL
BUN / creatinine	7–20 / 0.6–1.2 mg/dL
Platelets / INR	150–400 ×10 ⁹ /L / 0.8–1.2

READ AN ABG IN FOUR MOVES

1. pH: low is acidosis, high is alkalosis.
2. PaCO₂: moves opposite the pH in a respiratory problem.
3. HCO₃⁻: moves with the pH in a metabolic problem.
4. Compensation: compare the change against what is expected — a normal pH with abnormal values may be full compensation or a mixed disorder, so do not judge from pH alone.

PATTERN RECOGNITION

- Warm, vasodilated, low SVR with high cardiac output points to distributive shock.
- Cold, high SVR with low cardiac index suggests cardiogenic or hypovolaemic shock; high filling pressures favour cardiogenic, low filling pressures favour hypovolaemia.
- Low filling pressures with low output points to hypovolaemia.
- Equal, elevated filling pressures with muffled tones and pulsus paradoxus suggest tamponade.
- A rising lactate despite resuscitation signals persistent hypoperfusion and should prompt reassessment and escalation in context.
- Rising ICP: widening pulse pressure, bradycardia, irregular breathing.
- Peaked T waves, then a widened QRS, follow a rising potassium.

PROFESSIONAL CARING AND ETHICAL PRACTICE (20%)

- Synergy Model: match nurse competencies to patient characteristics.
- Advocacy and moral agency: speak for the patient's stated wishes.
- Clinical inquiry: question routine practice, use current evidence.
- Facilitation of learning: teach at the family's level and check understanding.
- Response to diversity: adapt care to values, language and beliefs.
- Systems thinking: use the chain of command and rapid-response resources.
- Autonomy, beneficence, non-maleficence, justice, veracity, fidelity.

TEST-DAY PLAN

1. About 72 seconds per item — flag and move.
2. Assess before you intervene, unless the stem shows an emergency.
3. Airway, breathing, circulation orders your priorities.
4. Choose the least invasive effective action first.
5. Answer every item; there is no penalty for a guess.

TIP Study aid only — not clinical guidance. Reference ranges vary by laboratory; confirm the current test plan in the AACN CCRN Exam Handbook.

